

A bioinformatics pipeline for processing and analysis of whole transcriptome sequence data

Ayam Gupta^{1,3}, Shalini Rajagopal^{1,3}, Sonal Gupta^{1,2,4},
Ashwani Kumar Mishra⁵ and Prashanth Suravajhala^{1,6*}

¹Department of Biotechnology and Bioinformatics, Birla Institute of Scientific Research, Jaipur 302001, Rajasthan, India; ²Department of Pediatrics, SMS Medical College and Hospital, Jaipur, Rajasthan, India; ³School of Biotechnology, Rajiv Gandhi Proudyogiki Vishwavidyalaya, State Technological University of Madhya Pradesh, Bhopal, Madhya Pradesh, India; ⁴Department of Biotechnology, Amity University Rajasthan, Kant Kalwar, Jaipur 303002, India; ⁵DNA Xperts, Noida, India, ⁶Bioclues.org, India

*For correspondence: prash@bisr.res.in

[Abstract]

Recent advances in next generation sequencing (NGS) technologies have paved the way for augmented genomic research. From the conventional microarray analysis to the current NGS-based whole transcriptome pipelines, there have been significant advances on how differentially expressed genes (DEGs) are identified. With several bioinformatics pipelines in use, inferring the candidate DEGs prove to be a cumbersome approach as one may have to reach consensus in analysing the results from various pipelines. To ensure this, we have benchmarked the well-known Cufflinks-Cuffdiff pipeline on a set of datasets and present it in the form of a protocol for the research community interested in performing downstream analysis of whole transcriptome shotgun sequencing.

Keywords: RNA-Seq, Transcriptome analysis, bioinformatics pipeline, differentially expressed genes

[Background]

Whole transcriptome shotgun sequencing (WTSS or RNA-Seq) allows us to discover new genes and transcripts and also measure their abundance in a single assay [1]. The quantitative expression of genes allows the researchers in predicting the molecular mechanisms underlying genome regulation. High depth RNA-Seq tends to be extremely sensitive allowing detection of lowly expressed transcripts when compared to microarray-based quantification [2–4]. This quantitative measurement of gene expression enables researchers to characterize transcripts from both coding and non-coding regions, identify abundances of RNA transcripts in diseased individuals and predict molecular mechanisms underlying genome regulation. Several robust and sensitive algorithms are available for the processing and analysis of RNA-Seq data. The protocols however vary based on how good the publicly available datasets were used for an efficient benchmarking. In this protocol, we present a bioinformatics workflow for quantitative analysis of RNA-Seq data and further compare it with publicly available computational tools using different algorithms. Our pipeline uses open-source tools to process and analyze RNA-Seq data starting with quality check until the last step of identification of differentially expressed genes (see Software).

Equipment

1. Computer

Ideal configuration: 32/64 GB RAM with 64 core CPUs in an Ubuntu operating system (16.04.4 LTS machine)

Software

All the softwares were downloaded from the following locations:

1. FastQC <https://www.bioinformatics.babraham.ac.uk/projects/FastQC>
2. Bowtie2 <https://bowtie-bio.sourceforge.net/Bowtie2>
3. HISAT2 <https://github.com/infphilo/Hisat2>
4. Samtools <http://samtools.sourceforge.net>
5. Cufflinks <https://github.com/cole-trapnell-lab/cufflinks>
6. DeSeq2 <https://bioconductor.org/packages/release/bioc/html/DESeq2.html>
7. FeatureCounts-apt-get install featurecounts
8. 1000 genomes dataset: <http://www.internationalgenome.org/>

Dataset Information

For benchmarking, we have downloaded publicly available 1000 genome datasets (<http://www.internationalgenome.org/>) with three different population accessions, viz. East Asian (ERR1050076 and ERR1050078), American (HG00731 and HG00733) and African ancestry (NA19238 and NA19240) (Table1). Each population contains an individual as parent and child and type of RNA sequencing is strand specific using Illumina platform. The files were downloaded either using FTP link or fasterq-dump (<https://www.internationalgenome.org/data-portal/sample>) [5].

Table 1. 1,000 genomes samples used for benchmarking

Population	Individual	Sample	Read length	Type	Technology	References
Han Chinese South, East Asian Ancestry	HG00512– Parent	ERR1050076	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/HG00512
Han Chinese South, East Asian Ancestry	HG00514– Child	ERR1050078	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/HG00514
Puerto Ricanin Puerto Rico, American Ancestry	HG00731– Parent	ERR1050079	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/HG00731
Puerto Ricanin Puerto Rico, American Ancestry	HG00733– Child	ERR1050081	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/HG00733
Yorubain Ibadan, Nigeria, African Ancestry	NA19238– Parent	ERR1050073	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/NA19238
Yorubain Ibadan, Nigeria, African Ancestry	NA19240– Child	ERR1050075	51	Strand specific RNA-Seq	Illumina	https://www.internationalgenome.org/data-portal/sample/NA19240

Procedure

The raw file (fastq) was processed using various steps which included quality check, alignment, sorting and annotation. The pipeline uses a combination of various tools, viz. FastQC, Bowtie2/HISAT2, samtools, cufflinks along with the human genome reference sequence (hg38 from UCSC human genome browser) and gene coordinates information file (Figure 1). Since the commands in pipeline run on Linux, all commands are case sensitive. This pipeline was run on a 64 GB RAM with 64 core CPUs in an Ubuntu operating system (16.04.4 LTS machine), but we recommend a minimum 16 GB RAM machine depending upon the raw fastq file size. A shell script was created with extension .sh and all the commands are provided below.

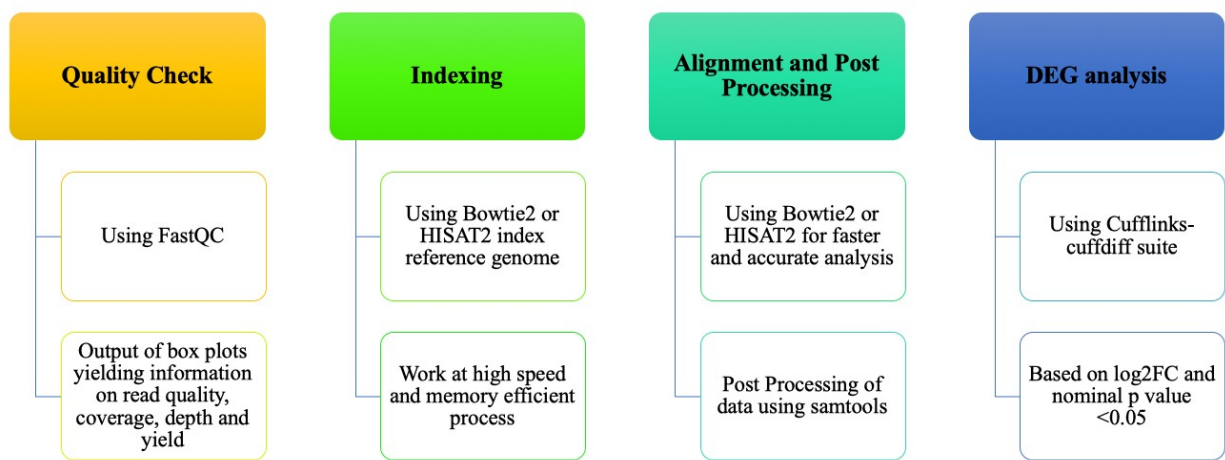


Figure 1: Flow chart for analysing RNA-Seq data.

A. Pre-processing of raw data

1. Quality check

The NGS analyses depend primarily on the raw data quality checks and so we used FastQC to plot the read depth and quality score besides performing a host of other statistical inferences (Figure 2).

```
./FastQC ~/samples/sample1.fastq
```

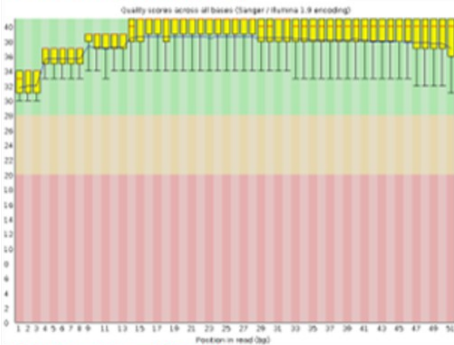
Command for quality check

Note: After a report is generated, several modules may be flagged as “Passed”, “Warn” or “Failed”. But solely relying on the flags is highly discouraged as the statistics change with the type of input data.

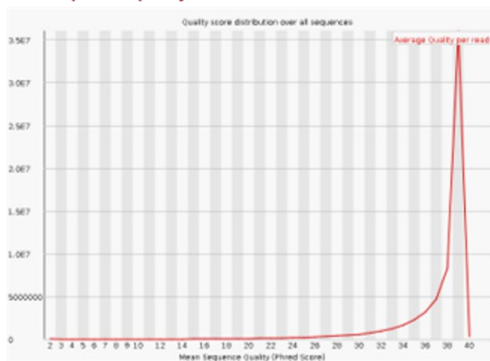
Basic Statistics

Measure	Value
Filename	ERR1050078_2.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	63949386
Sequences flagged as poor quality	0
Sequence length	51
%GC	48

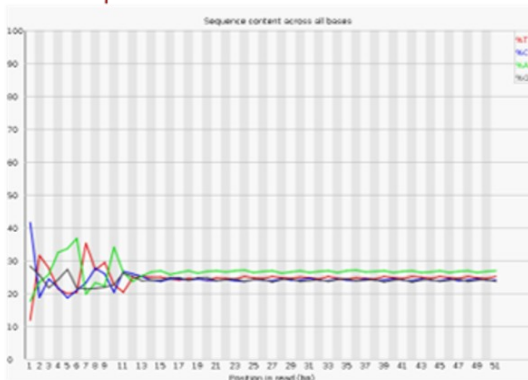
Per base sequence quality



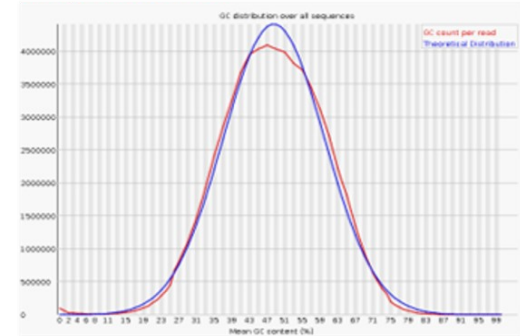
Per sequence quality scores



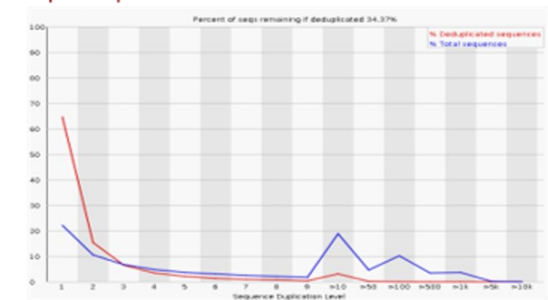
Per base sequence content



Per sequence GC content



Sequence Duplication Levels



Kmer Content

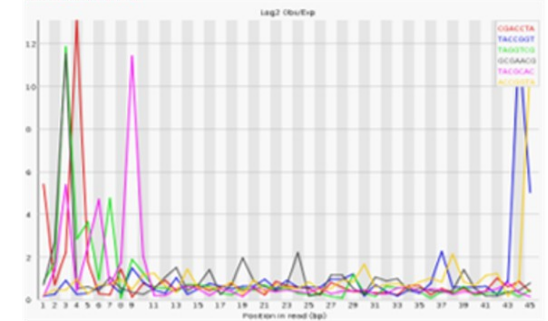


Figure 2: Summary plots obtained from a standard FastQC run, containing information on statistics, quality, read coverage, depth, yield, based per read call, *etc*

From the FASTQC results, per sequence quality score plots the total number of reads vs the average quality score over full length reads. The Phred score of above 25 is considered good wherein from the y

axis, the quality calls range from very good (green), the calls of reasonable quality (orange) can be measured, while in pink, they are considered worse. The reads that fall in those regions could be trimmed using cutadapt which is beyond the scope of this protocol. The distribution of average read quality should be fairly tight in the upper range of the plot, whereas sequence content plot reports the percent of bases called for each of the four nucleotides at each position across all reads in the file. Kmer content plot measures the count of each short nucleotide of length k (default = 7) starting at each position along the read.

2. Indexing human genome

Bowtie2 and HISAT2 were used to index reference genomes both of which work at high speed and are memory efficient. Compared to Bowtie2, the Hisat2 indexing is more memory efficient although the latter may skip intron-exon splice junctions.

```
./bowtie2 --build reference_inbt2_index_base
./hisat2 --build reference_inht2_index_base
#builds indices which are required for aligning reads
```

When the commands are run, Hisat2-build outputs a set of 8 files (with suffixes .1.ht2, .2.ht2, .3.ht2, .4.ht2, .5.ht2, .6.ht2, .7.ht2, and .8.ht2). These files together constitute the index: that is needed to align reads to that reference. The original sequence FASTA files are no longer used by Hisat2, once the index is built. In case of Bowtie2, the current directory would contain six new files that all start with reference and end with .1.bt2, .2.bt2, .3.bt2, .4.bt2, .rev.1.bt2, and .rev.2.bt2. While the first four files are the forward strands, the rev files indicate the reverse indexed sequences in case of Bowtie2.

3. Alignment and post processing

Hisat2 is a fast and sensitive alignment used to map RNA-Seq reads to the genome. Like Bowtie2, Hisat2 also uses burrows-wheeler transform (BWT) to compress the genomes so that they require very little memory to store. Unlike BWA and Bowtie, Hisat2 builds a whole genome global index and thousands of small local indexes to make spliced alignment possible to run on a standard system in an efficient way [6]. Bowtie2 is used for short read alignment and uses very little RAM with accuracy and modest performance in indexing the alignment [7]. The mismatch or any sequencing errors or small genetic variation between samples and reference genome could be checked using the following commands:

```
./hisat2 -x reference_hisat_filename -1 path/filename1 -2 path/filename2 > filename.sam
./bowtie2 -x reference_filename -1 path/filename1 -2 path/filename2 > filename.sam
```

Note: The -2 option may be omitted for single-end sequences and -U may be used instead.

HISAT2 and Bowtie2 align a set of unpaired reads (in fastq or .fq format) to the reference genome using the Ferragina and Manzini (FM) – index [7]. The alignment results output in SAM format [8] and a short alignment summary is written to the console.

Samtools is a collection of tools to manipulate the resulting alignment in SAM/BAM format. SAM stands for sequence alignment/map format and its corresponding format is binary mapped format (BAM). SAM could be converted into different alignment format, sort, merge, alignment, remove duplicates, call SNPs and short indel variants; whereas the BAM and following indices (.bai) are used to view the index of the binary aligned sequences. The basic necessity of having the binary mapped files is to save the memory.

```
./samtools import reference.fa ~/samples/sample1.sam > sample1.bam
```

4. **Sorting BAM**

A sorted BAM file is used for streamlined data processing to avoid extra alignment when loaded into memory. Whereas it can be easily indexed for faster retrieval of alignment, overlapping is needed to retrieve alignment from a region. All BAM files need to be sorted and indexed for better performance. Samtools sort command is used to convert the BAM file to assorted BAM file and samtools index to index BAM file.

```
~/samtools sort ~/samples/sample1.bam -o sample1.sorted  
~/samtools index ~/samples/sample1.sorted
```

B. **Examining differentially expressed genes**

The Cufflinks suite of tools is used to perform a number of analyses for RNA-Seq experiments. It assembles the transcripts and estimates their abundances, and finally examines differentially expressed genes (DEG) in RNA-Seq samples. The Cufflinks suite includes several different programs that work together to perform these analyses.

1. `./cufflinks in.sorted.bam -o sample.out`

It is included in the Cufflinks package which helps in merging the assemblies. The merged assembly provides us a basis for calculating genes and transcripts expression in each condition which are finally analysed for DEGs through Cuffdiff command.

2. `./cuffmerge -g reference.gtf -s reference.fa sample.merged`

Cuffmerge would merge all the replicate samples into one, while Cuffdiff compares the aligned reads from RNA-Seq samples from two or more conditions and identifies transcripts that are differentially expressed using a thorough statistical analysis, viz. the log₂ fold change (log₂FC) which is then reported as a set of

text files plotted graphically.

```
3. ./Cuffdiff ~/samples/sample1.out/transcripts.gtf ~/samples/sample1.sorted ~/samples/reference_genome
> sample_result
```

The current protocol is focused on the Cufflinks-Cuffdiff pipeline wherein the results are compared to the already established DESeq2 pipeline using standard cutoff parameters. For the benchmarking process, two different sequence aligners, i.e., Bowtie2 and Hisat2 were used for concision and comparison. Standard cutoff parameters were considered for the analysis, i.e., log2FC and nominal p value < 0.05

Feature counts is a program for assigning mapping reads to genomic features such as genes, exons, promoters and genomic bins [9].

```
Feature Counts -B -p -g gene_id -a ~/reference.gtf -o ~/sample.sorted.bam ~/sample.sorted.bam
```

DESeq2 is used for differential gene expression analysis based on the negative binomial distribution. It is a Bioconductor package which is continually updated and available in R.

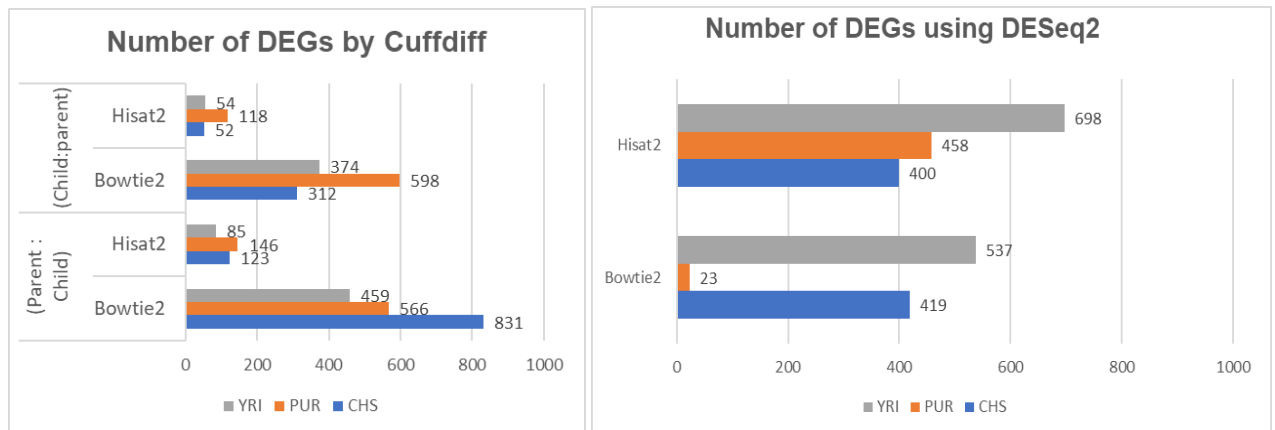
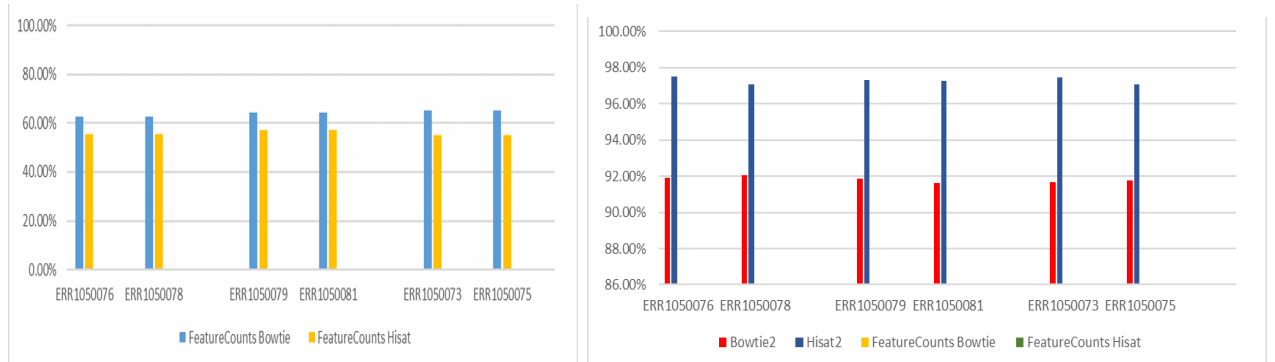
Data Analysis

1. Alignment of sequencing reads

Benchmarking yielded a good recovery rate for DEGs through RNA-Seq analysis. From six samples, we obtained an RNA-Seq quality checked using FastQC. The reads were aligned to the human reference genome assembly (hg38) that were downloaded from NCBI GenBank (ftp link) and then reaching consensus with two aligning tools, *viz.* Bowtie2 and Hisat2. The average alignment percentage was found to be 91.82% and 97.27% for Bowtie2 and Hisat2 respectively with the percentage of reads mapped to the reference genome found to be similar between all six groups suggesting that there was no sequencing bias in the data (Table2). Further from coverage percentage analysis between DESeq2 pipeline and the pipeline we used, this is observed as much lower than Bowtie2 (~90%) and Hisat2 (>90%) generated alignment files from our pipeline across all the samples (Figure 3) as unannotated regions are also covered by it. The Bowtie2 and Hisat2 generated bam files were then used to generate the read counts files for DESeq2 analysis.

Table 2. Coverage obtained using Bowtie2 and Hisat2 aligners

SAMPLE	BOWTIE2	HISAT2
ERR1050076	91.91%	97.49%
ERR1050078	92.07%	97.05%
ERR1050079	91.87%	97.29%
ERR1050081	91.64%	97.27%
ERR1050073	91.66%	97.45%
ERR1050075	91.75%	97.09%



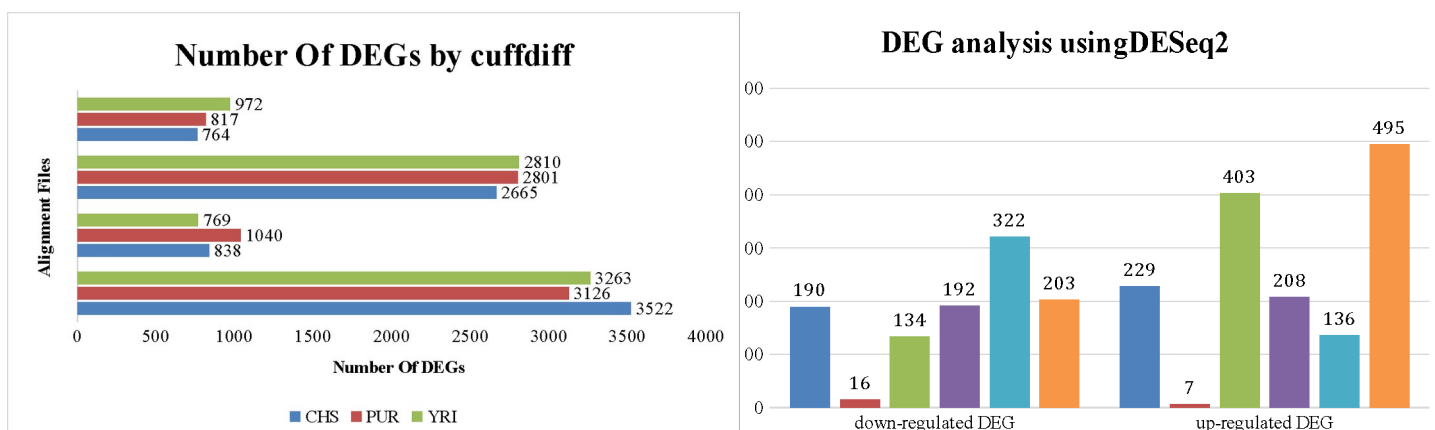
(a)

(b)

Figure 3: (a) comparison of coverage between Bowtie2 and Hisat2. (b) Number of DEGs aligned by Hisat2 and Bowtie2 and the total number of DEGs identified using the DESeq2 pipeline.

2. Differentially expressed genes

The files were processed using cufflinks software with alternating the transcripts.outfile from both the samples of the same population (Figure3). Cuffdiff analysis was done using transcripts.outfile (output file from Cuffdiff) with parent:child as well as child:parent relationships which yielded diverse results.



We obtained an average of 272 (YRI), 356 (PUR) and 477 (CHS) DEGs using Cuffdiff taken as parent:child and while taken as child:parent, we obtained an average of 214 (YRI), 358 (PUR) and 182 (CHS) DEGs. However, from the DESeq2 pipeline, we obtained a **higher** number of average DEGs among different populations as compared to our pipeline (Figure 3b and 3c). The up- and down-regulated gene counts obtained from the DESeq2 pipeline in different populations were plotted in the form of bar graphs (Figure 4) and Volcano plots, generated by comparing DEGs obtained from both Bowtie2 and Hisat2 (Figure 5 and 6). While there is no general objective justification for any particular log2FC threshold, from a mathematical point of view, it is possible to reject the null hypothesis at any non-zero log2FC if the variability is low enough. We have set a log2FC and p-values and q-values < 0.05, as log2FC is robustly detected from RNA-Seq and qPCR. The DESeq2 code can be accessed via https://github.com/ayamgupta/RNAseq_benchmark.

Figure 4: Down-regulated and up-regulated gene count using the Cuffdiff and DESeq2 pipeline

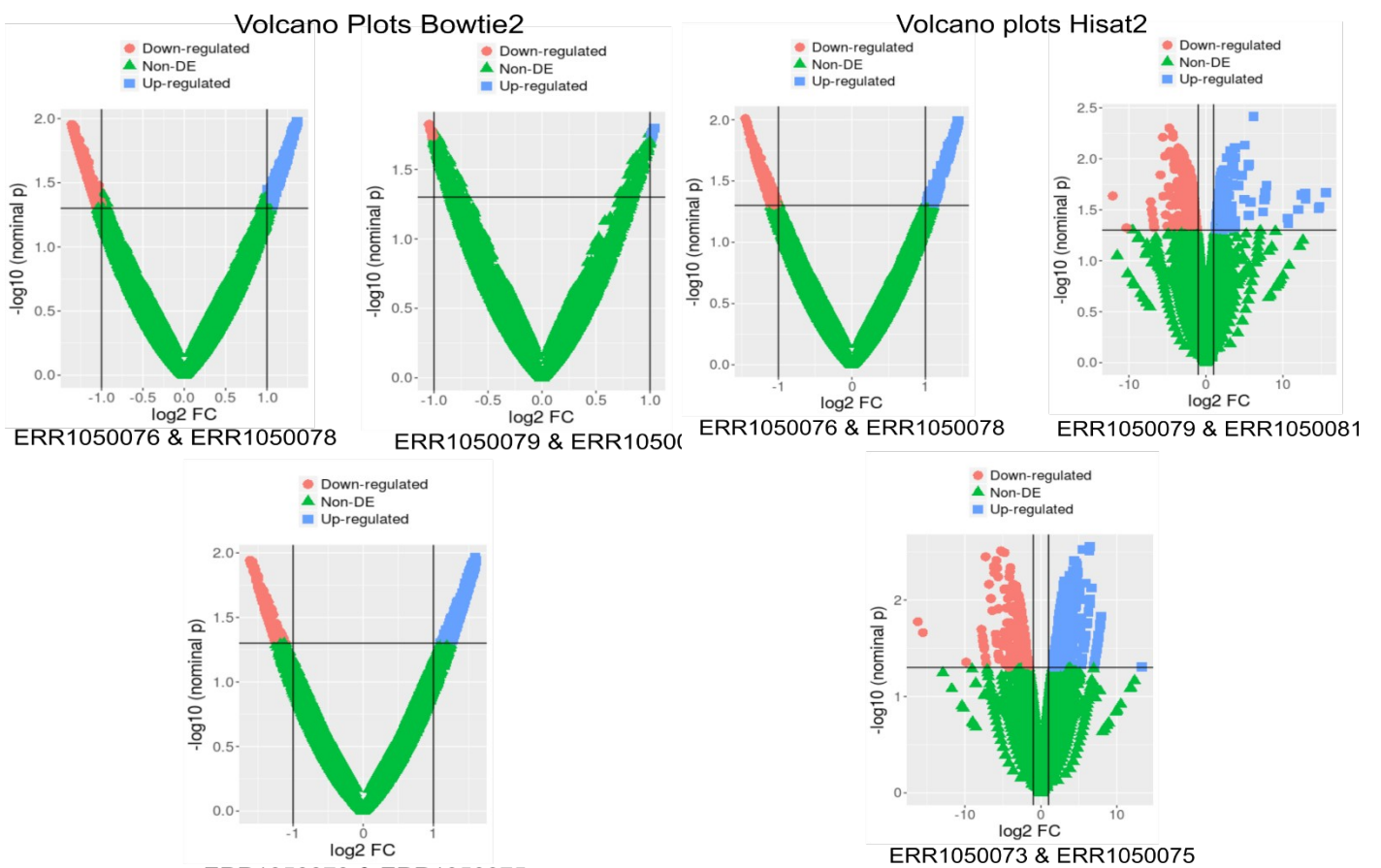
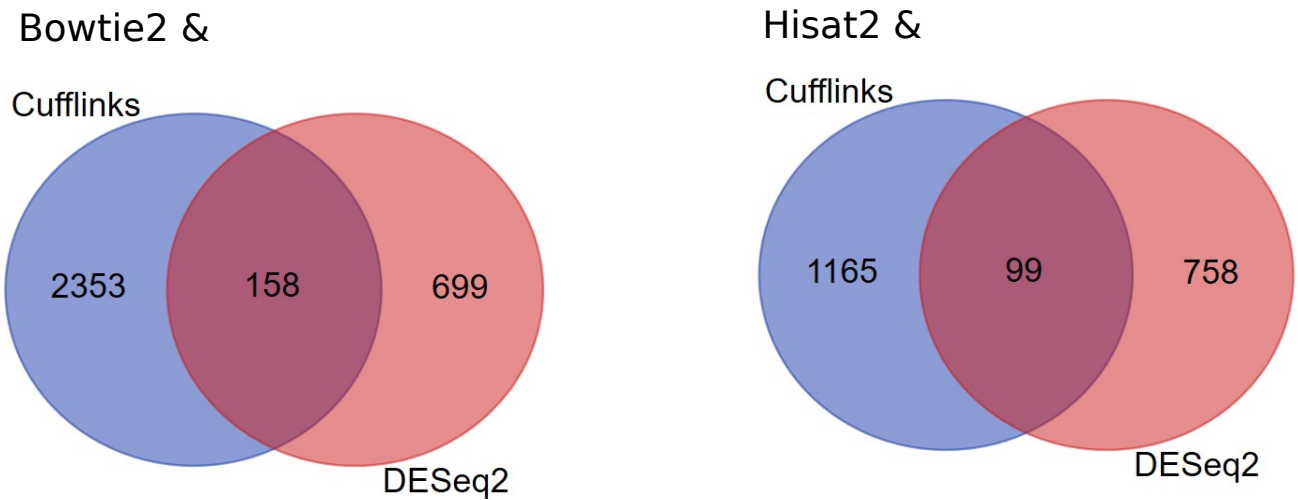


Figure 5: Volcano plots from DESeq2 analysis showing log2 fold (DEGs) change in the samples aligned using (a) Bowtie2 and (b) Hisat2



Samples	Down-regulated DEG	Non DEG	Up-regulated DEG
ERR1050076 & ERR1050078	190	57976	229
ERR1050079 & ERR1050081	16	58372	7
ERR1050073 & ERR1050075	134	57858	403
ERR1050076 & ERR1050078	192	57995	208
ERR1050079 & ERR1050081	322	57937	136
ERR1050073 & ERR1050075	203	57697	495

Figure 6: From the downstream processing of files, 158 common genes were observed between the Bowtie2- Cufflinks pipeline and Deseq2 in which 99 genes were observed to be common across Hisat2- Cufflinks pipeline. A significant difference in the number of discovered DEGs was found in comparing the Cufflinks-Cuffdiff pipeline when compared to DESeq2. These differences can be attributed to the coverage of unannotated regions by the latter to some extent.

Conclusions

Benchmarking is a gold standard for any NGS pipeline. It gives us inherent statistical interpretation on how the data is used for further analyses. In this protocol, we have presented a RNA-Seq pipeline which can be aptly used by beginners. While the Cufflinks-Cuffdiff pipeline is a simplest approach to identify the DEGs, the tool such as DESeq is another alternative. It is unto the discretion of the end-user who should weigh the approaches and finally reach consensus before analyzing the results.

References

- [1] Trapnell C, Roberts A, Goff L, Pertea G, Kim D, Kelley DR, et al. Differential gene and transcript expression analysis of RNA-seq experiments with TopHat and Cufflinks. Nat Protoc 2012. <https://doi.org/10.1038/nprot.2012.016>.
- [2] Wang C, Gong B, Bushel PR, Thierry-Mieg J, Thierry-Mieg D, Xu J, et al. The concordance between

- RNA-seq and microarray data depends on chemical treatment and transcript abundance. *Nat Biotechnol* 2014. <https://doi.org/10.1038/nbt.3001>.
- [3] Nookaew I, Papini M, Pornputtapong N, Scalcinati G, Fagerberg L, Uhlén M, et al. A comprehensive comparison of RNA-Seq-based transcriptome analysis from reads to differential gene expression and cross-comparison with microarrays: A case study in *Saccharomyces cerevisiae*. *Nucleic Acids Res* 2012. <https://doi.org/10.1093/nar/gks804>.
 - [4] Zhao S, Fung-Leung WP, Bittner A, Ngo K, Liu X. Comparison of RNA-Seq and microarray in transcriptome profiling of activated T cells. *PLoS One* 2014. <https://doi.org/10.1371/journal.pone.0078644>.
 - [5] Altshuler DL, Durbin RM, Abecasis GR, Bentley DR, Chakravarti A, Clark AG, et al. A map of human genome variation from population-scale sequencing. *Nature* 2010. <https://doi.org/10.1038/nature09534>.
 - [6] Kim D, Langmead B, Salzberg SL. HISAT: A fast spliced aligner with low memory requirements. *Nat Methods* 2015. <https://doi.org/10.1038/nmeth.3317>.
 - [7] Langmead B, Salzberg SL. Fast gapped-read alignment with Bowtie 2. *Nat Methods* 2012. <https://doi.org/10.1038/nmeth.1923>.
 - [8] Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, et al. The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 2009. <https://doi.org/10.1093/bioinformatics/btp352>.
 - [9] Liao Y, Smyth GK, Shi W. FeatureCounts: An efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* 2014. <https://doi.org/10.1093/bioinformatics/btt656>.